The mixture form of the Version-4 basket model: model, likelihood, sampling and training

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Abstract

This document states the model completely and derives everything that is computed from it. Section 1 fixes notation. Section 2 defines the model: a prior over a latent taste z, an exact law of the basket given z, the marginal law of the basket, and the posterior of z given a basket. Section 3 shows how the law given z is normalised exactly, and Section 4 how a basket is drawn from it exactly, as a chain of three conditional draws. Section 5 explains how the likelihood of an observed basket is estimated, and Section 6 how the parameters are trained. Section 7 covers generation and counterfactuals, Section 8 compares with the earlier form of the model, and Section 9 gives the evidence so far.

Contents

1 Notation **2**

2 The model **2**
2.1 Parameters . . . 2
2.2 Context utility . . . 2
2.3 Prior . . . 3
2.4 The law of the basket given the taste . . . 3
2.5 The marginal law: the model . . . 4
2.6 The joint law and the posterior . . . 4

3 Computing the conditional law exactly **4**

4 Sampling **5**
4.1 A basket given a taste: three conditional draws . . . 5
4.2 A basket from the model . . . 6

5 The likelihood of an observed basket **6**
5.1 The posterior has one peak . . . 6
5.2 The proposal . . . 7
5.3 The estimator . . . 7

6 Training **8**
6.1 Objective . . . 8
6.2 Gradient . . . 8
6.3 One training step . . . 8
6.4 Moment matching . . . 8
6.5 Whitening (parameter-expanded EM) . . . 9
6.6 Validation . . . 9

7 Generation and counterfactuals **9**

8 Relation to the earlier form **10**

9 Evidence so far **10**

10 Open points **11**

1 Notation

- A *trip* t is one store visit by household $h = h(t)$ at store $s(t)$ in week $w(t)$. Its context x_t holds prices, discounts, display and mailer flags, the week, the store and the household's purchase history before the trip.
- $\mathcal{J}_t$ is the set of products offered on trip t. Every product j belongs to one category $c(j) \in \{1, \dots, C\}$; $\mathcal{J}_{tc} = \{j \in \mathcal{J}_t : c(j) = c\}$. On the Dunnhumby commodity data $C = 187$ and there are 5,455 products.
- A *basket* is a non-empty set $S \subseteq \mathcal{J}_t$. We write

$$n_c(S) = |S \cap \mathcal{J}_{tc}|, \qquad n(S) = \sum_c n_c(S), \qquad m(S) = \#\{c : n_c(S) > 0\}$$

  for the number of products in category c, the basket size and the number of categories present.
- Support: $\mathcal{S}_t = \{S \subseteq \mathcal{J}_t : \ 1 \le n(S) \le n_{\max},\ n_c(S) \le R\ \forall c\}$ with $n_{\max} = 120$ and $R = 40$. All laws below are on $\mathcal{S}_t$.
- r is the taste dimension ($r = 16$). $\mathcal{N}(z; \mu, \Sigma)$ is the Gaussian density on $\mathbb{R}^r$. $(a)_+ = \max(a, 0)$, $\binom{k}{2} = k(k-1)/2$.

The trip index t is dropped when only one trip is involved.

2 The model

2.1 Parameters

symbol	per	meaning	trained?
λ_j	product	popularity intercept	yes
$\phi_j \in \mathbb{R}^r$	product	taste loading of product j	yes
$\psi_c \in \mathbb{R}^r$	category	taste loading of an *extra* product of category c	yes
ρ_c	category	penalty per pair of products within c	yes
$f_c(2), f_c(3), f_c(\ge 4)$	category	free penalty on holding 2, 3, ≥ 4 products of c	yes
$\rho_0(1..n_{\max})$	count	size potential on m, the number of categories present	yes
rest of b_{tj}		context utility, (1)	yes*

The fit (`scripts/fit_mixture.py`) starts from the exact additive base model ($\phi = 0$) and trains every parameter except the within-category competition weights in b_{tj}, which the price instrument re-estimates once per epoch. *The real-data fit of Section 9 is older: it started from the earlier form's checkpoint `runs/commodity/v4_comp_promo.pt` and kept the rest of b_{tj} at its values.

2.2 Context utility

For product j on trip t,

$$\begin{aligned} b_{tj} = \lambda_j + \theta_h \cdot \alpha_j + \theta_h \cdot u_{c(j)} - g_{hj}\left[\bar{p}_t + \kappa(p_{tj} - \bar{p}_t)\right] - g_{hj}(\kappa_d - \kappa)\, d_{tj} - \gamma_d\, d_{tj} \\ + w_j^{\mathrm{dsp}} D_{tj} + w_j^{\mathrm{mlr}} M_{tj} + \mu_j \cdot \delta_{w(t)} + \zeta_j \cdot \xi_{s(t)} + \eta_t(j) - o_t + \iota_{t,c(j)}. \end{aligned} \tag{1}$$

$\theta_h \cdot \alpha_j$ and $\theta_h \cdot u_c$ are household tastes (product and category); p_{tj} is the log price deviation, $\bar{p}_t$ its mean over the trip's assortment, d_{tj} the discount depth, $g_{hj} \geq 0$ the household-by-product price sensitivity, $\kappa, \kappa_d > 0$ the relative-price and discount multipliers, γ_d a common discount slope; D, M are display and mailer flags; $\mu_j \cdot \delta_w$ and $\zeta_j \cdot \xi_s$ are season and store terms; $\eta_t(j)$ is the purchase-history term; o_t is a trip-level offset (the mean of η_t over the assortment plus a history-size term), the same for every product of the trip; and

$$\iota_{tc} = (\ell_c - 1)\big[\mathrm{IV}_{tc} - \mathrm{IV}_{tc}^{\mathrm{reg}}\big], \qquad \mathrm{IV}_{tc} = \log \sum_{j \in \mathcal{J}_{tc}} e^{b^{\circ}_{tj}}, \quad \ell_c \in (0,1),$$

is the within-category competition shift (b° is (1) without ι and o_t, and "reg" the same with price deviation, discount, display and mailer set to zero). With ℓ_c near 0 a price or promotion change in c moves purchases between products of c rather than raising the category's purchase rate. In this document b_{tj} is simply a known number per (trip, product): it does not depend on z or on the basket.

2.3 Prior

Each trip has a latent taste

$$z_t \sim \mathcal{N}(0, I_r), \qquad \text{independently across trips.} \tag{2}$$

z_t describes what the shopper is in the mood for on this trip (for example "baking" or "baby products"). It is not tied to the household: two trips of the same household draw independent tastes; persistent household preferences are in θ_h inside b_{tj}.

2.4 The law of the basket given the taste

Definition 1 (conditional law)**.** *Given z, the probability of basket $S \in \mathcal{S}_t$ is*

$$p(S \mid z) = \frac{\exp E_z(S)}{Z_z}, \qquad Z_z = \sum_{S' \in \mathcal{S}_t} \exp E_z(S'), \tag{3}$$

with

$$E_z(S) = \sum_{j \in S} u_j(z) - \sum_c h_c\big(n_c(S), z\big) - \rho_0\big(m(S)\big), \tag{4}$$

$$u_j(z) = b_j - \tfrac{1}{2}\|\phi_j\|^2 + z \cdot \phi_j, \tag{5}$$

$$h_c(k, z) = \rho_c\tbinom{k}{2} + f_c(k) + (k-1)_+\big(z \cdot \psi_c - \tfrac{1}{2}\|\psi_c\|^2\big), \qquad h_c(0, z) = 0. \tag{6}$$

Here $f_c(0) = f_c(1) = 0$, $f_c(k) = f_c(\geq 4)$ for $k \geq 4$, and $\rho_0(1) = 0$ fixes the free constant.

Reading (4): each product contributes its utility $u_j(z)$; a taste z raises the products whose loading ϕ_j points the same way. Within a category, every additional product beyond the first pays ρ_c per pair and f_c, and its taste contribution is corrected by ψ_c (a category that is liked need not be bought many times). The size term charges the number of categories present.

The basket field. Collect the z-dependence of $E_z(S)$:

$$E_z(S) = a(S) + z \cdot s(S), \qquad s(S) = \sum_{j \in S} \phi_j - \sum_c (n_c(S) - 1)_+ \, \psi_c \in \mathbb{R}^r, \tag{7}$$

$$a(S) = \sum_{j \in S} \big(b_j - \tfrac{1}{2}\|\phi_j\|^2\big) - \sum_c \Big[\rho_c\tbinom{n_c}{2} + f_c(n_c) - \tfrac{1}{2}(n_c - 1)_+\|\psi_c\|^2\Big] - \rho_0(m).$$

$a(S)$ does not involve z. Equation (7), *the energy is affine in z*, is the one fact every property below uses. $s(S)$ is called the basket field: the taste direction the basket points in.

Why $-\frac{1}{2}\|\phi_j\|^2$? It makes the expected boost of product j under the prior zero on the multiplicative scale: $\mathbb{E}_{z\sim\mathcal{N}(0,I)} e^{z\cdot\phi_j - \frac{1}{2}\|\phi_j\|^2} = 1$. It is a choice of parameterisation (it could be absorbed into λ_j), kept so that the earlier form and the mixture form share their parameters.

2.5 The marginal law: the model

Definition 2 (mixture form)**.** *The probability that trip t produces basket S is*

$$p(S) = \int_{\mathbb{R}^r} \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz. \tag{8}$$

This is a continuous mixture: one basket law $p(\cdot \mid z)$ for every taste z, weighted by the prior density of that taste. Equivalently, the data are generated hierarchically:

$$z_t \sim \mathcal{N}(0, I), \qquad S_t \mid z_t \sim p(\cdot \mid z_t).$$

Products bought together are explained by a shared taste. The number of components is not a choice: there is one per z, they all share the formula (3), and the parameter count does not depend on it. The modelling choice is the dimension r and the Gaussian shape of the prior. $p(S)$ is a proper probability: $\sum_S p(S) = \int \mathcal{N}(z) \sum_S p(S \mid z)\, dz = 1$.

2.6 The joint law and the posterior

The joint density of taste and basket is $p(z, S) = \mathcal{N}(z; 0, I)\, p(S \mid z)$. By Bayes' rule the posterior of the taste given an observed basket is

$$p(z \mid S) = \frac{\mathcal{N}(z; 0, I)\, p(S \mid z)}{p(S)}, \qquad \log p(z \mid S) = -\tfrac{1}{2}\|z\|^2 + z \cdot s(S) - \log Z_z + \text{const}(S), \tag{9}$$

where the second form uses (7); the constant collects $a(S)$, $\log p(S)$ and the Gaussian constant. The posterior is the main computational object: the likelihood $p(S)$ is its normaliser, and training averages over it.

3 Computing the conditional law exactly

Z_z is a sum over all baskets, astronomically many. It is computed exactly because, given z, the weight $e^{E_z(S)}$ factorises over categories except for the size term, which depends on one integer.

Elementary symmetric polynomials. Write $w_j = e^{u_j(z)}$. For category c and $k \geq 0$,

$$e_{c,k} = \sum_{A \subseteq \mathcal{J}_c,\ |A| = k} \prod_{j \in A} w_j,$$

the total product weight of all ways to choose k products from c. These follow from the recursion over the products of c in a fixed order: $e_k^{(p)} = e_k^{(p-1)} + w_p\, e_{k-1}^{(p-1)}$, $e_0^{(0)} = 1$.

Category polynomials. The total weight of category c holding exactly k products is

$$G_c(k) = e_{c,k}\, e^{-h_c(k,z)}, \qquad G_c(0) = 1. \tag{10}$$

The size table. Let $F_c(n,m)$ be the total weight of all choices in categories $1,\dots,c$ that hold n products in m present categories:

$$F_0(n,m) = [n=0, m=0], \qquad F_c(n,m) = F_{c-1}(n,m) + \sum_{k=1}^{\min(R,n)} F_{c-1}(n-k, m-1)\, G_c(k). \tag{11}$$

Equivalently $F_C(n,m)$ is the coefficient of $t^n y^m$ in $\prod_c \big(1 + y \sum_{k\ge 1} G_c(k) t^k\big)$.

Proposition 1 (exact normaliser)**.**

$$Z_z = \sum_{n=1}^{n_{\max}} \sum_{m=1}^{n} F_C(n,m)\, e^{-\rho_0(m)}. \tag{12}$$

Proof. A basket is a choice, for each category, of a subset of size $n_c \le R$. Its weight $e^{E_z(S)}$ is $e^{-\rho_0(m)} \prod_c \big(\prod_{j\in S_c} w_j\big) e^{-h_c(n_c,z)}$. Summing over the subsets with the counts (n_c) fixed gives $e^{-\rho_0(m)} \prod_c G_c(n_c)$; (11) sums $\prod_c G_c(n_c)$ over all count vectors with total n and m non-zero entries. □

The cost is $O(\sum_c |\mathcal{J}_c| R)$ for the polynomials and $O(C\, n_{\max}^2 R)$ for the table, per trip and per z. Numerically every row n of the table carries its own power-of-two exponent, because weights of different sizes differ by far more than a double can hold. The gradients follow by differentiating (12):

$$\nabla_z \log Z_z = \mathbb{E}[s(S) \mid z], \qquad \nabla_z^2 \log Z_z = \mathrm{Cov}[s(S) \mid z], \qquad \frac{\partial \log Z_z}{\partial u_j} = P(j \in S \mid z), \tag{13}$$

the first two because $\log Z_z = \log \sum_S e^{a(S) + z\cdot s(S)}$. All three are exact, computed by reverse-mode differentiation through (10)–(12). Likewise, for any model parameter ϑ, $\nabla_\vartheta \log Z_z = \mathbb{E}_{S\sim p(\cdot|z)}[\nabla_\vartheta E_z(S)]$.

4 Sampling

4.1 A basket given a taste: three conditional draws

Given z, a basket is drawn exactly from (3) by the following chain of conditional draws, using the tables of Section 3.

Step A. Size and number of categories. Draw (n,m) with

$$P(n,m \mid z) = \frac{F_C(n,m)\, e^{-\rho_0(m)}}{Z_z}. \tag{14}$$

Step B. Count per category, going backwards $c = C, C-1, \dots, 1$. Let (n', m') be the products and categories still to place in categories $1..c$ (initially (n,m)). Draw $n_c = k$ with

$$P(n_c = k \mid n', m', z) = \frac{F_{c-1}\big(n'-k,\ m' - [k>0]\big)\, G_c(k)}{F_c(n',m')}, \qquad 0 \le k \le \min(R, n'), \tag{15}$$

then set $n' \leftarrow n' - k$, $m' \leftarrow m' - [k>0]$. The probabilities sum to one by (11), and at the end $n' = m' = 0$.

Step C. Products within each category. For every category with $n_c = k > 0$, draw a subset A of $\mathcal{J}_c$ of size k with

$$P(A \mid n_c = k, z) = \frac{\prod_{j\in A} w_j}{e_{c,k}}. \tag{16}$$

This is done sequentially: go through the products $p = L, L-1, \ldots, 1$ of the category with q still to choose, and include p with probability $w_p\, e_{q-1}^{(p-1)} / e_q^{(p)}$ (after which $q \leftarrow q-1$).

Proposition 2 (exactness)**.** *The basket produced by Steps A–C has law $p(S \mid z)$.*

Proof. Multiply the probabilities. In Step B the denominators telescope against the numerators of the next category, and Step A's numerator cancels the first denominator, so

$$P(n,m)\prod_c P(n_c \mid \cdot) = \frac{e^{-\rho_0(m)}}{Z_z}\, F_0(0,0)\prod_c G_c(n_c) = \frac{e^{-\rho_0(m)}}{Z_z}\prod_c e_{c,n_c} e^{-h_c(n_c,z)}.$$

In Step C the inclusion probabilities telescope in the same way, because $e_q^{(p)} = e_q^{(p-1)} + w_p e_{q-1}^{(p-1)}$, and their product is (16). Multiplying by $\prod_c \prod_{j\in A_c} w_j / e_{c,n_c}$ gives

$$\frac{e^{-\rho_0(m)}}{Z_z}\prod_c e^{-h_c(n_c,z)}\prod_{j\in S} w_j = \frac{e^{E_z(S)}}{Z_z}.$$

□

Sampling is therefore hierarchical: size and categories first, then the count in each category given what is left, then the products given the count. Each conditional is the exact conditional of $p(S \mid z)$.

4.2 A basket from the model

Proposition 3 (exact draws from the model)**.** *Draw $z \sim \mathcal{N}(0, I)$, then S by Steps A–C given z. Then $S \sim p(S)$ of (8).*

Proof. The pair (z, S) has density $\mathcal{N}(z; 0, I)\, p(S \mid z)$; its S-marginal is (8). □

No Markov chain is involved, so there is no burn-in, no mixing question and no correlation between draws. The full hierarchy of a generated basket is therefore

$$z \;\to\; (n, m) \;\to\; (n_1, \ldots, n_C) \;\to\; \text{products}.$$

Truncation for speed. The sampler first builds the table only up to a per-trip cap $K < n_{\max}$ chosen so that baskets larger than K carry at most 10^{-8} of Z_z (a proven bound); if the bound is not met the table is rebuilt to $n_{\max}$. The draw is exact up to that 10^{-8}.

5 The likelihood of an observed basket

$p(S)$ in (8) is an $r = 16$-dimensional integral with no closed form. It is estimated by importance sampling over z, which works well because the integrand has one peak.

5.1 The posterior has one peak

Proposition 4 (strong log-concavity)**.** *For every basket S, $\log p(z \mid S)$ is 1-strongly concave in z:*

$$\nabla_z \log p(z \mid S) = s(S) - z - \mathbb{E}[s \mid z], \qquad \nabla_z^2 \log p(z \mid S) = -\big(I + \mathrm{Cov}[s \mid z]\big) \preceq -I.$$

Hence it has exactly one maximiser $\hat{z}(S)$, the solution of $\hat{z} = s(S) - \mathbb{E}[s \mid \hat{z}]$, and $\log p(z \mid S) \le \log p(\hat{z} \mid S) - \frac{1}{2}\|z - \hat{z}\|^2$ for all z.

Proof. Differentiate (9) using (13). A covariance is positive semi-definite, so the Hessian is at most $-I$. A strongly concave function has a unique maximiser, and the quadratic bound is Taylor's theorem with the Hessian bound. □

Finding the peak. Gradient ascent $z \leftarrow z + \eta \nabla_z \log p(z \mid S)$, with the step $\eta \le 1$ halved until $\log p(z \mid S)$ rises by at least $\frac{1}{4}\eta\|\nabla\|^2$ (Armijo), so every iteration increases it. Each iteration needs $\mathbb{E}[s \mid z]$, one exact gradient of $\log Z_z$. Validation runs 30 iterations from $z = 0$; training runs 8 starting from the trip's peak at its previous visit.

5.2 The proposal

Around the peak the posterior is close to the Gaussian with covariance equal to the inverse negative Hessian (Laplace approximation),

$$\Sigma(S) = \big(I + \mathrm{Cov}[s \mid \hat{z}]\big)^{-1},$$

where $\mathrm{Cov}[s \mid \hat{z}]$ is the sample covariance of $s(S')$ over 32 exact draws $S' \sim p(\cdot \mid \hat{z})$ (Section 4). The proposal is

$$q(z \mid S) = 0.8\,\mathcal{N}\big(z; \hat{z},\ 1.2^2\Sigma(S)\big) + 0.2\,\mathcal{N}\big(z; \hat{z},\ I\big). \tag{17}$$

The first part follows the posterior's shape, slightly widened; the second has unit covariance, which by Proposition 4 is at least as wide as the posterior in every direction.

5.3 The estimator

Draw $z_1, \ldots, z_K \sim q(\cdot \mid S)$ independently and set

$$\omega_k = \frac{\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)}{q(z_k \mid S)}, \qquad \hat{p}_K(S) = \frac{1}{K}\sum_{k=1}^{K} \omega_k. \tag{18}$$

Each $p(S \mid z_k)$ is exact by (3) and (12).

Proposition 5 (properties of the estimate)**.** *1.* $\mathbb{E}\,\hat{p}_K(S) = p(S)$ *(unbiased).*
2. $\mathbb{E}\log\hat{p}_K(S) \le \mathbb{E}\log\hat{p}_{K+1}(S) \le \log p(S)$*, and* $\mathbb{E}\log\hat{p}_K(S) \to \log p(S)$ *as* $K \to \infty$*.*
3. Every weight is bounded: $\omega_k \le (2\pi)^{r/2}\, p(S)\, p(\hat{z} \mid S)/0.2$*.*

Proof. (1) $\mathbb{E}_q[\mathcal{N}p(S \mid z)/q] = \int \mathcal{N}(z)p(S \mid z)\,dz$. (2) The upper bound is Jensen's inequality; monotonicity in K and convergence are the importance-weighted bound of Burda, Grosse and Salakhutdinov (2016). (3) $\mathcal{N}(z; 0, I)p(S \mid z) = p(S)p(z \mid S) \le p(S)p(\hat{z} \mid S)e^{-\frac{1}{2}\|z-\hat{z}\|^2}$ by Proposition 4, and $q(z) \ge 0.2\,(2\pi)^{-r/2}e^{-\frac{1}{2}\|z-\hat{z}\|^2}$. □

So the reported log-likelihood can understate the model but on average never overstates it, and no single draw can dominate without bound. The effective sample size

$$\mathrm{ESS} = \frac{(\sum_k \omega_k)^2}{K\sum_k \omega_k^2} \in (0, 1]$$

is reported as a quality check. Validation uses $K = 256$.

Posterior averages. The normalised weights $\bar{\omega}_k = \omega_k / \sum_l \omega_l$ estimate averages over the posterior, $\mathbb{E}[g(z) \mid S] \approx \sum_k \bar{\omega}_k g(z_k)$. This estimate is consistent as $K \to \infty$ but biased for small K: it leans towards the proposal. Section 6.5 shows where this matters.

6 Training

6.1 Objective

Let ϑ be the trained parameters (Section 2.1) and $t = 1, \dots, N$ the training trips (N = 158,983). The target is the penalised log-likelihood

$$\frac{1}{N}\sum_{t=1}^{N} \log p_\vartheta(S_t) - \frac{1}{N}\Pi(\vartheta), \qquad \Pi = \tfrac{1}{2}\Big[20\sum_c \|\psi_c\|^2 + 200\sum_j \|\phi_j - \psi_{c(j)}\|^2 + 0.2\sum_c \rho_c^2 + 2\sum_{c,i} f_{c,i}^2\Big],$$

a Gaussian prior that keeps each product's loading near its category's, plus the Gaussian priors on the rest of b_{tj}: price parameters centred at the base model's values, the household taste per category and the history terms at zero, and a pooling penalty on the three product-by-context factorisations (`v4/training.py`). Since $\log p_\vartheta(S_t)$ is not available, training maximises the K-sample bound of Proposition 5,

$$\mathcal{L}_K(\vartheta) = \frac{1}{N}\sum_t \mathbb{E}_{z_{t,1..K}\sim q_t}\Big[\log \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{N}(z_{tk}; 0, I)\, p_\vartheta(S_t \mid z_{tk})}{q_t(z_{tk})}\Big] - \frac{\Pi(\vartheta)}{N}, \tag{19}$$

with $K = 8$. The proposal q_t is built from the current parameters but treated as fixed when differentiating: the bound holds for any proposal, so this is legitimate.

6.2 Gradient

For fixed draws, $\nabla_\vartheta \log \hat{p}_K(S) = \sum_k \bar{\omega}_k \nabla_\vartheta \log p_\vartheta(S \mid z_k)$, and

$$\nabla_\vartheta \log p_\vartheta(S \mid z) = \nabla_\vartheta E_z(S) - \mathbb{E}_{S'\sim p(\cdot\mid z)}\big[\nabla_\vartheta E_z(S')\big]. \tag{20}$$

The first term is the observed basket's; the second is the model's expectation given the same taste, exact by reverse-mode differentiation through the dynamic program of Section 3. No Markov chain and no model samples enter the gradient. The gradient is the posterior-weighted average of "observed minus expected given the taste".

6.3 One training step

For a batch of 256 trips:

1. Find each trip's posterior peak $\hat{z}_t$ (8 Armijo iterations, warm-started).
2. Draw 32 baskets $S' \sim p(\cdot \mid \hat{z}_t)$ exactly and form $\Sigma(S_t)$ and the proposal (17).
3. Draw $z_{t1}, \dots, z_{t8} \sim q_t$.
4. Compute $\log \hat{p}_8(S_t)$ with gradients, add the penalty, and take one Adam step (gradient norm clipped at 10). Learning rates: 3×10^{-3} for ϕ, ψ; 0.05, 0.03, 0.006, 0.008 for ρ_0, ρ_c, f, λ.
5. Apply the bounds: $\rho_c \geq -1.5$ and the floors on f_c and the category reward.

Every 30 steps two extra updates follow (Sections 6.4 and 6.5). An epoch is 622 steps; the fit runs 2 epochs.

6.4 Moment matching

At a maximum of the likelihood the gradient in λ_j vanishes, and by (20) that gradient is "observed rate of j minus model rate of j" (posterior-weighted). The same holds for $\rho_0(m)$ (frequency of m categories present) and ρ_c (mean $\binom{n_c}{2}$). Adam approaches this slowly for rarely bought products, so every 30 steps each of these parameters takes the step

$$\lambda_j \leftarrow \lambda_j + 0.35\,\mathrm{clip}\Big(\frac{\text{data rate}_j - \text{model rate}_j}{\text{data rate}_j}, \pm 1\Big),$$

with the rates averaged over the last 30 steps (and factors 0.5 for ρ_0, ρ_c). For small errors (data − model)/data ≈ log(data/model), which is the change in λ_j that makes a Poisson rate match. This changes the speed of convergence, not its target.

6.5 Whitening (parameter-expanded EM)

The idea. Suppose the prior were $\mathcal{N}(0, \Sigma_0)$ with Σ_0 a free parameter. The EM update of Σ_0 is the average posterior second moment

$$\Sigma_0^{\text{new}} = \frac{1}{N} \sum_t \mathbb{E}\big[zz^\top \mid S_t\big]. \tag{21}$$

A model with prior $\mathcal{N}(0, \Sigma_0)$ is the same law as one with prior $\mathcal{N}(0, I)$ and loadings rescaled: put $z = A\tilde{z}$ with $A = \Sigma_0^{1/2}$ (symmetric), $\tilde{z} \sim \mathcal{N}(0, I)$. Then $z \cdot \phi_j = \tilde{z} \cdot A\phi_j$ and $z \cdot \psi_c = \tilde{z} \cdot A\psi_c$. So each whitening step sets

$$\phi_j \leftarrow A\phi_j, \qquad \psi_c \leftarrow A\psi_c, \qquad \lambda_j \leftarrow \lambda_j + \tfrac{1}{2}\big(\|A\phi_j\|^2 - \|\phi_j\|^2\big),$$

the last so that u_j in (5) is unchanged. This is an EM step for the scale and orientation of the tastes, which gradient steps on ϕ alone find slowly. If the model is right, $\Sigma_0^{\text{new}} = I$ and the step does nothing.

Not exactly the same law. The term $\frac{1}{2}(n_c - 1)_+\|\psi_c\|^2$ in (6) becomes $\frac{1}{2}(n_c - 1)_+\|A\psi_c\|^2$ and is not compensated, so the step is an approximate EM step.

Estimating (21)**.** The posterior second moments are estimated with normalised importance weights (Section 5.3) on 256 fixed training trips with $K = 256$. An earlier version reused the training draws ($K = 8$); that estimate is biased upward, because with few draws the normalised weights lean toward the proposal, which is wider than the posterior. Measured on the same 1,024 trips and the same model:

estimate of $\frac{1}{rN}\operatorname{tr}\sum_t \mathbb{E}[zz^\top \mid S_t]$	$K = 8$	$K = 256$
value (1 = model correctly scaled)	1.056	0.990

With $K = 8$ every whitening enlarged the tastes by about 3% that the data did not ask for; over many rounds generated baskets grew from 7.66 to 8.23 products (data 7.26). With $K = 256$ this stops.

6.6 Validation

Every 120 steps: $\log \hat{p}_{256}$ on 64 held-out trips (the fit's validation split; test weeks are not used), and the size distribution of exact draws (Section 4.2) for 1,024 held-out trips against the data. The checkpoint with the best validation log-likelihood is kept.

7 Generation and counterfactuals

Generation. For trip t: build b_{tj} from the context, draw $z \sim \mathcal{N}(0, I)$, then Steps A–C. Repeating gives the distribution of baskets for that trip. Incidence has an exact inner part:

$$P(j \in S) = \mathbb{E}_{z \sim \mathcal{N}(0,I)}\big[P(j \in S \mid z)\big], \qquad P(j \in S \mid z) = \partial \log Z_z / \partial u_j \text{ from (13)},$$

so only the outer average over z is Monte Carlo.

Counterfactuals. A price or promotion change alters b_{tj} (and ι_{tc}) only. Draw z once, then draw S under the old and the new context with the same z: the same shopper's taste before and after the change. In this form the distribution of tastes is fixed by the prior and does not respond to prices; prices change what each taste buys. Effects on whether a trip happens at all are outside the model.

Conditioning on part of a basket. For recommendations given a partly observed basket T, average over $p(z \mid T \subseteq S) \propto \mathcal{N}(z)P(T \subseteq S \mid z)$. Unlike Proposition 4 this posterior is not guaranteed to have one peak, because $\log P(T \subseteq S \mid z)$ is a difference of two convex functions.

8 Relation to the earlier form

The earlier Version-4 law was $p_{\text{old}}(S) = e^{a(S)+\frac{1}{2}\|s(S)\|^2}/Z$ with the same a and s. Using $e^{\frac{1}{2}\|s\|^2} = \int \mathcal{N}(z;0,I)e^{z\cdot s}dz$,

$$p_{\text{old}}(S) = \int \underbrace{\frac{\mathcal{N}(z;0,I)\,Z_z}{Z}}_{\text{weight of taste } z}\; p(S \mid z)\,dz.$$

Both forms use the same $p(S \mid z)$; they differ in the weight of a taste. In the earlier form a taste is weighted by $\mathcal{N}(z)Z_z$, and Z_z grows enormously for tastes that make large baskets attractive. Its logarithm $-\frac{1}{2}\|z\|^2 + \log Z_z$ is concave plus convex and can have several peaks. On the Dunnhumby refit one peak, far from the data, made 120-product baskets carry most of the probability. Training (Gibbs chains started at the data) and validation (annealed chains) both missed it: the reported held-out log-likelihood was -38.8 per trip, and -53.8 once that peak was included. In the mixture form the weight is $\mathcal{N}(z)$ alone, the posterior has one peak (Proposition 4), draws are exact, and the likelihood estimate is a lower bound. The two forms are different laws with the same parameters, so one can start the fit of the other.

9 Evidence so far

Exactness. In the test world (6 products, 3 categories, $r = 2$) and all 11 model variants, $\log p(S \mid z)$ from (3)–(12) equals enumeration of every basket to 10^{-9}, and the estimate (18) with 4,000 draws equals enumeration plus 48×48 Gauss–Hermite quadrature of (8) to within 0.01 nats (`tests/test_law.py`).

Synthetic data (48 categories, 522 products). Held-out log-likelihood -27.38 against -27.90 for the earlier form fully trained; mean generated size 7.92 against 8.69 in the data.

Real data (64 held-out trips). The earlier form's old fit scores -36.78 with a peak-covering estimate. The mixture form, in the full fit started from it:

step	log-lik.	mean ESS	size, mean	size, 99%
data			7.26	42.8
240	-35.777	0.35	7.66	41
360	$\mathbf{-35.744}$	0.31	7.86	44.8
480 (whitening, $K = 8$)	-35.895	0.27	8.23	46

The run was restarted from step 360 with the whitening estimate of Section 6.5; no generated basket has reached the 120 cap at any point.

10 Open points

- The whitening step is approximate (Section 6.5).
- The trained bound (19) with $K = 8$ is below $\log p$; how much it biases the fitted parameters is measured only indirectly (validation at $K = 256$).
- In the real-data fit of Section 9 only $\lambda, \phi, \psi, \rho_0, \rho_c, f$ were trained; price, promotion, competition and history terms kept the earlier fit's values. The fit from the additive base model, which trains them all, has not been run yet, and the price and promotion checks have to be re-run on this form.
- The prior is one Gaussian. A mixture of a few Gaussians would keep Proposition 4 within each component.